

Problem 3

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Problem 1 Determine the value of a constant b for which f is continuous at 0. **EXPLAIN.**

$$f(x) = \begin{cases} \frac{2x+b}{x-5} & \text{if } x < 0 \\ \frac{x+16}{x^2-16} & \text{if } x \geq 0 \end{cases}$$

Explanation. In order for f to be continuous at $x = 0$, the three conditions of continuity have to be satisfied. These criteria are:

$$\begin{aligned} &f \text{ is defined at } x = 0 \\ &\lim_{x \rightarrow 0} f(x) \text{ exists} \\ &\lim_{x \rightarrow 0} f(x) = f(0) \end{aligned}$$

The first condition is satisfied. In order to satisfy the second condition, $\lim_{x \rightarrow 0} f(x)$ exists, we must verify that $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$. That is,

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \left(\frac{x+16}{x^2-16} \right) = \frac{16}{-16} = -1 \\ \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{2x+b}{x-5} = \frac{b}{-5} \\ \implies \lim_{x \rightarrow 0} f(x) &= -1 = \frac{-b}{5} \implies b = 5 \end{aligned}$$

and $f(0) = -1$. f is continuous at $x = 0$ for $b = 5$ since $\lim_{x \rightarrow 0} f(x) = f(0) = -1$.

Notice here, it's not enough to just find a value for b . Our work must also justify the continuity at $x = 0$ for that value of b .